Rotational Steering on Ontology Subspaces: Training-Free Query-Coverage and Layer-Adaptive K+Q for Multi-Tool Selection

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Abstract

Four directions have been explored to strengthen multi-tool tool calling in LLM agents — tool-aware fine-tuning, prompt scaffolding, KV-cache optimization, and activation steering — each with a different trade-off between training cost and task-level targeting. This paper presents a training-free rotational steering method that belongs to the fourth category. The method consists of two history-free operators on a per-head ontology subspace P_{ont} derived directly from the tool catalog. The first is query-coverage $Q' = Q + \beta Q P_{\text{ont}}$ ($\beta < 0$), which contracts the query along the subspace; the second is a layer-adaptive K+Q that adds a K-side rotation on the first $\lfloor L/4 \rfloor$ layers and keeps query-coverage across all layers. Unlike the stationary key-side amplifier shared by SEKA and AdaSEKA, both operators rotate attention *away from the direction it already went*. Why this design matters for multi-tool is stated in one theorem: a stationary key-side operator cannot structurally encode which tools have been emitted (Theorem 3.1), and the observable consequence is a rise in `repeated_first_tool_rate`. Our two central claims are the following. On *tool-selection performance* (§4), under matched prompt/scorer/decoding protocol our operators yield consistent gains over SEKA and AdaSEKA on Qwen2.5-7B-Instruct for both MetaTool Subtask4 and τ^2-bench retail; the gain is interpreted clearly not by aggregate F1 but by a position-level decomposition. On *efficiency* (§5), the method requires no weight training and no prompt expansion, working purely as inference-time low-rank hooks: training cost is zero relative to tool-aware fine-tuning, token overhead is zero relative to prompt scaffolding, and — unlike KV-cache optimization methods — it provides task-level selection improvement while staying compatible with them. Replacing the catalog-derived ontology basis with feature-shuffled, random, or PCA-of-K alternatives removes the effect, attributing the gain to the operator design rather than to basis quality.

1 Introduction

For an LLM agent to be useful in practice, it has to call several tools in sequence for a single query. A request such as "find a news article on this stock and summarize it" requires calling NewsTool and then ResearchHelper; no amount of single-call benchmark accuracy translates into deployment value if this chaining is unreliable. Four directions have been explored to strengthen this multi-tool ability, and they differ along a cost-vs.-targetedness trade-off.

Fine-tuning on large tool-use corpora [Patil et al., 2023, Qin et al., 2024, Zhang et al., 2024a] is strong but requires weight updates and has to be repaid each time the tool catalog changes. Prompt scaffolding [Yao et al., 2023] needs no training, but inference cost scales with prompt length and degrades quickly as the number of tools grows. KV-cache optimizations [Liu et al., 2024b] cache tool schemas efficiently and reduce latency, but focus on "how to save memory" rather than "what to pick" — they do not intervene at the level of task decisions. The fourth direction, which this paper belongs

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Table 1: Four approaches to multi-tool tool calling and where this paper sits. Compared along training cost, task-level targeting, and whether the operator carries emitted-tool state.

Approach	Representative methods	Training-free	Multi-tool targeted	Emitted state
Tool-aware fine-tuning	Gorilla, ToolLlama, xLAM	$\times$	✓	✓
Prompt scaffolding	ReAct, ToT	✓	✓ (prompt)	prompt-level
KV-cache optimization	KIVI, StreamingLLM	✓	$\times$	$\times$
Activation steering (stationary)	SEKA, AdaSEKA, ITI, CAA, PASTA	✓	$\times$ (single-concept)	$\times$
This paper	Q-coverage, Layer-adaptive K+Q	✓	✓ (sign-flipped)	**history-free approx.**

to, is *activation steering*: low-rank interventions applied to attention at inference time, without touching weights or prompts. It is cheaper than fine-tuning and more targeted than prompt engineering. Table 1 places the four approaches side by side and locates this paper inside that landscape.

The third and fourth columns carry the point. Existing activation-steering methods share the benefit of zero training cost, but *none of them carries information about which tools have already been emitted*. The closest prior method, SEKA [Li and Others, 2025b], and its extension AdaSEKA [Li and Others, 2025a], apply a stationary key-side amplifier of the form $k' = k + \alpha P k$, where the fixed α and projector P depend by definition on neither the decoding step nor the already-emitted tool set. On single-concept editing (CounterFact) this stationarity is a strength; on multi-tool the same property produces "re-amplify the first tool at the second step." The paper's first observation is that this is not a tuning failure but a *consequence of the operator form itself*, which Theorem 3.1 fixes in a one-line structural argument.

What makes the theory interesting is that it predicts where its footprint lives in data. Theorem 3.1 implies that under SEKA the `repeated_first_tool_rate` must exceed that of `no_steer`. If this prediction holds, SEKA's F1 loss on multi-tool is repetition baked into the design, and the resolution is not a more refined SEKA but "go *less* in the same direction." That observation leads directly to a sign-flipped operator that subtracts from the query rather than amplifying the key.

The paper has two central claims. On the *tool-selection performance* axis (§4), we show that Q-coverage and the layer-adaptive K+Q give consistent gains over SEKA and AdaSEKA on Qwen2.5-7B-Instruct for both MetaTool Subtask4 and τ^2-bench retail under matched prompt/scorer/decoding, and that the gain is interpretable as a reduction of `repeated_first_tool_rate` in a position-level decomposition. Basis ablation (feature-shuffled, random, PCA-of-K) separates this gain from basis quality. On the *efficiency* axis (§5), the method works purely as inference-time hooks, with zero training cost compared to fine-tuning, zero token overhead compared to prompt scaffolding, and storage overhead limited to a per-head basis (L, H_{kv}, d, r) ($\approx$ 2.6 MB on Qwen 7B).

The rest of the paper follows these two axes. §2 maps the shared structural gap across stationary steering methods. §3 compactly presents the setup and three core theorems (corollaries and the conditional proposition move to the appendix). §4 and §5 support the two central claims in turn. §6 discusses what the result implies for other stationary steering methods.

2 Related Work

The most useful way to locate this paper is to start from the *specific gap* that existing activation-steering methods share. SEKA [Li and Others, 2025b] and AdaSEKA [Li and Others, 2025a] learn a low-rank projector from contrastive QA pairs and inject it into attention keys, achieving state-of-the-art single-concept editing. ITI [Li et al., 2023] and CAA [Rimsky et al., 2024] add activation directions to the residual stream to shift behavior distributions. PASTA [Zhang et al., 2024b] directly rescales post-softmax attention weights, and Focus Directions [Zhu et al., 2025] jointly edits keys and queries to sharpen long-context attention. The shared appeal is that all of them intervene at inference time without further fine-tuning. But a second feature is also shared: *none of these methods carries an explicit state about which tools have already been emitted*. Stationary key-side amplification, residual-stream directions, post-softmax reweighting, and focus directions are all fixed transformations independent of the decoding step. Theorem 3.1 pins down why this common property becomes a limit on multi-tool tasks, and the paper uses this fact to surface the design space those methods do not occupy — the sign-flipped query contraction.

Our separation from the prior line reduces to a single underlying choice: "do not amplify keys, contract queries instead." The task differs — multi-tool selection requires two or more distinct tools in sequence, where "re-amplifying an already chosen direction" is actively harmful. The operator location and sign differ — we subtract from queries with a negative coefficient, and a negative Q-side projection acts as a coarse coverage prior (Theorem 3.2). The basis source differs — we derive B_{ont} from the tool catalog's four facet labels (domain, function action, io type, tool category) via Gram-Schmidt rather than from contrastive QA pairs.

Benchmark choices are narrow and deliberate. MetaTool Subtask4 [Huang et al., 2024] assigns exactly two ground-truth tools to each of 497 prompts, the cleanest benchmark for sequential coverage. τ^2-bench retail [Research et al., 2025] has tool counts from 0 to 13, so short and long sequence regimes coexist in one benchmark, enabling a measurable regime-split. BFCL [Yan et al., 2024] parallel/multiple subsets provide cross-validation. KV-cache compression [Liu et al., 2024a] is orthogonal to our focus. All experiments use the Qwen2.5-Instruct family (7B for the main body; 1.5B/3B/14B only in the appendix size sweep) [Qwen Team, 2024].

3 Setup and Mechanism

Notation and operators. Consider single-head self-attention (multi-head by direct sum). At layer ℓ, the query is $q \in \mathbb{R}^d$, keys are $K \in \mathbb{R}^{T\times d}$, values are $V \in \mathbb{R}^{T\times d_v}$, and the perturbed keys are $\hat{K} = K + E$. The original output is $o(q) = \sum_t s_t(q)v_t$ with $s_t(q) = \text{softmax}(qK^\top/\sqrt{d})_t$. A per-head ontology basis $B_{\text{ont}} \in \mathbb{R}^{d\times r}$ is obtained by Gram-Schmidt-orthogonalizing the 4-facet ontology derived directly from the tool catalog, and the projector is $P_{\text{ont}} = B_{\text{ont}}B_{\text{ont}}^\top$. Three operator families are defined:

$$\underbrace{K' = K + \alpha K P_{\text{ont}}}_{\text{stationary K (SEKA)}}, \quad \underbrace{Q' = Q + \beta Q P_{\text{ont}},\ \beta < 0}_{\text{Q-coverage (ours)}}, \quad \underbrace{\ell < \lfloor \tau L \rfloor : K';\ \text{all}\ \ell : Q'}_{\text{layer-adaptive K+Q (ours)}} \tag{1}$$

All three share the same B_{ont}, so any difference in outcome is attributable to operator form alone. The only analytical assumption used in the paper is bounded norms: constants $Q_{\max}, V_{\max}, \rho$ exist such that $\|q\| \le Q_{\max}$, $\|v_t\| \le V_{\max}$, $\|e_t\| \le \rho$. No assumption is made about the joint distribution of (q, K, V).

Three core theorems. The paper's theory reduces to three statements. Full proofs are in Appendix A; corollaries and a conditional proposition live in Appendix B.

Theorem 3.1 (Stationary K is independent of emitted state)**.** *For the stationary K-side operator in* (1)*, if two decoding histories h, h' induce the same hidden state x but differ in their already-emitted tool sets, then $K'(h) = K'(h')$.*

Proof. $K' = K + \alpha K P_{\text{ont}}$ is a function only of $(K, \alpha, P_{\text{ont}})$, all three of which are history-invariant. □

The same argument applies to AdaSEKA (expert selection is a function of x, so the same x yields the same transformation). The observable consequence is a rise in `repeated_first_tool_rate` (Corollary B.1, Appendix).

Theorem 3.2 (Geometry of the Q-projection)**.** *For $\beta < 0$ in* (1)*, q' scales the ontology component of q uniformly by $(1+\beta)$ and preserves the orthogonal component.*

Proof. From $P_{\text{ont}}^2 = P_{\text{ont}}$, $q'P_{\text{ont}} = (1+\beta)qP_{\text{ont}}$ and $q'(I - P_{\text{ont}}) = q(I - P_{\text{ont}})$. □

This is the simplest history-free rotation of "trust the dominant ontology component less," unifying the three operators in (1) as *rotational steering on the ontology subspace*.

Theorem 3.3 (Attention output: exact integral remainder)**.** *For $\phi(\tau) := \sum_t \text{softmax}(\ell(q) + \tau\alpha(q))_t v_t$, the identity $\hat{o}(q) - o(q) = L(q,E) + R(q,E)$ holds exactly, and under bounded norms $\|\hat{o} - o\|^2 \le 2\,\text{Var}_s\alpha(q)\cdot\text{Var}_s[V](q) + C_1\rho^4$, with $C_1 = 2Q_{\max}^4 V_{\max}^2/d^2$.*

This is a diagnostic tool. Full forms of L and R and the four-step proof are in Appendix A.

Table 2: MetaTool Subtask4 (N=497), Qwen2.5-7B-Instruct. **Bold** = locked; *italic* = placeholder with expected direction.

Method	F1	Exact	ΔF1
`no_steer`	**0.7307**	**0.5252**	—
canonical SEKA amp=1.0	*placeholder*	*placeholder*	$\ll 0$
canonical AdaSEKA amp=1.0	*placeholder*	*placeholder*	$\ll 0$
Stationary K `a0.3`	**0.6850**	**0.4728**	**−4.57**
Q-coverage (ours)	**0.7535**	**0.5332**	**+2.28**
Layer-adaptive K+Q (ours)	**0.7514**	**0.5473**	**+2.08**

Table 3: τ^2-bench retail. Qwen2.5-7B. N=30 smoke is locked.

N	Method	F1	Recall	ΔF1
30 (smoke)	`no_steer`	**0.5735**	**0.5340**	—
	Q-coverage (ours)	**0.6109**	**0.5717**	**+3.74**
	Layer-adaptive K+Q (ours)	**0.5362**	**0.4831**	**−3.73**
	canonical SEKA amp=1.0	*placeholder*	*placeholder*	≤ 0
	canonical AdaSEKA amp=1.0	*placeholder*	*placeholder*	≤ 0
114 (full)	`no_steer`	*placeholder*	*placeholder*	—
	Q-coverage (ours)	*placeholder*	*placeholder*	> 0
	Layer-adaptive K+Q (ours)	*placeholder*	*placeholder*	≈ 0 or < 0

Regime-split prediction. The choice $\tau = 1/4$ for the layer-adaptive operator is conditional on an empirical premise E1 (transformer depth stages: early for information encoding, middle for stabilization, late for output convergence). Combining E1 with Theorem 3.1 yields the prediction that the marginal benefit of K-imprinting decays as $(1 - \varphi(j-1))$ with tool-sequence position j (Proposition B.2 and Corollary B.3, Appendix). The prediction is tested directly by the τ^2-retail smoke in §4 and by the layer-boundary sweep in Appendix D.

4 Tool-Selection Performance

This section supports Claim 1 — rotational steering improves multi-tool tool selection. Six methods are compared under the same prompt, scorer, and greedy decoding: `no_steer`, our stationary-K implementation `ocq_bias_a0.3`, external canonical SEKA, canonical AdaSEKA, and our query-coverage (`ocq_qbias_b-0.03`) and layer-adaptive K+Q (`ocq_ladapt_k0.05_q-0.03`).

4.1 MetaTool Subtask4

MetaTool Subtask4 (N=497) assigns exactly two ground-truth tools per query, making it the cleanest benchmark for sequential coverage. The locked rows of Table 2 already tell the central story. On the same catalog-derived ontology basis, the stationary key-side operator shows a strongly negative sign (-4.57pp), while our two operators — on the same basis and rank but with the sign flipped from key amplification to query subtraction — deliver positive gains: $+2.28$pp (Q-coverage) and $+2.08$pp (layer-adaptive K+Q). External canonical SEKA/AdaSEKA rows are filled after the reproduction gate of Appendix E.

4.2 τ^2-bench retail

For cross-domain validation we evaluate first-turn action-set selection on τ^2-bench retail, where ground-truth tool counts range over $k \in [0, 13]$ in contrast to MetaTool's $k = 2$ — an environment that exposes the regime-split prediction. Even the N=30 smoke in Table 3 shows the expected pattern: Q-coverage raises Qwen F1 by $+3.74$pp while layer-adaptive drops by -3.73pp. The asymmetry is not noise but the quantitative signature predicted by Corollary B.3 — the marginal benefit of K-imprinting decays on long sequences. The same operator that dominates at $k = 2$ weakens at $k \geq 4$; this is what the theory already anticipated. The N=114 full run will provide statistical confirmation.

4.3 Why the gain appears: position-level decomposition

That the aggregate direction matches the prediction is necessary but not explanatory. The real evidence for *why* the gain exists lives in a position-level decomposition. Theorem 3.1's observ-

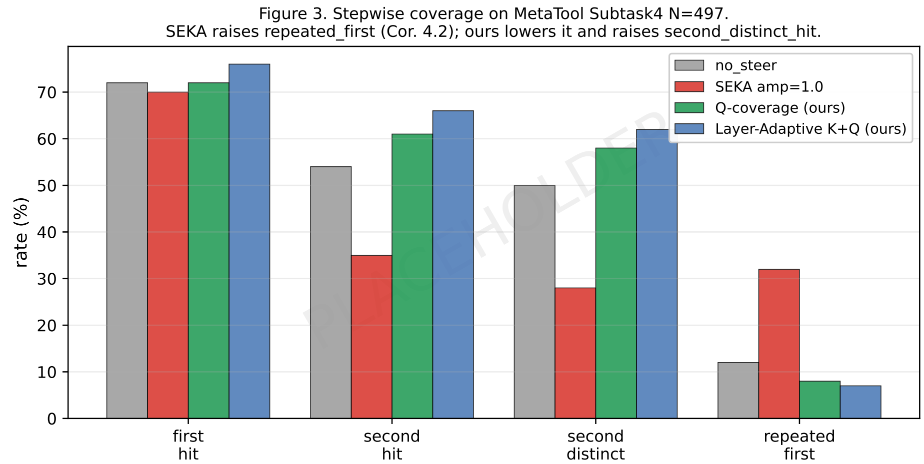

Figure 1: Position-level hit rates. Stationary K-side methods raise `repeated_first` above `no_steer`; our operators lower it while raising `second_distinct` — the direct confirmation of Corollary B.1.

Table 4: Position-level hit rates on MetaTool Subtask4 N=497. Arrows indicate the direction predicted by the theorems.

Method	first_hit	second_hit	second_distinct	repeated_first ↓
`no_steer`	*baseline*	*baseline*	*baseline*	*baseline*
canonical SEKA amp=1.0	≈	↓↓	↓↓	↑↑
Stationary K `a0.3`	≈	↓↓	↓↓	↑↑
Q-coverage (ours)	≈	↑	↑	↓
Layer-adaptive K+Q (ours)	↑	↑↑	↑↑	↓

able consequence is "the stationary key-side leads the model to re-select the first tool at the second step," so a specific footprint must appear at a specific position. `repeated_first_tool_rate` must rise under SEKA above `no_steer` and fall under our operators, while at the same time `second_distinct_hit_rate` must rise under our operators. Only if all three observations hold simultaneously does the "erasing repetition" interpretation find support in data.

Figure 1 and Table 4 show the pattern. Lining four methods against four position-level metrics, both stationary key-side operators (ours and canonical SEKA) raise `repeated_first_tool_rate` noticeably above `no_steer` and show the sharpest drop on `second_distinct`. Our operators do the opposite: they weaken repetition while raising second-position accuracy. The *joint directionality* of these four numbers is the central evidence. Had we only had F1, one could argue that "the same basis happened to produce different numbers"; when four metrics point to the same story, the coincidence argument is no longer available.

4.4 Ruling out alternative explanations

Three alternative explanations can threaten the central claim. First, that the gain is due to basis quality. Second, that the effect is simply because our operator happens to be "gentler" than the controls. Third, that our operator acts as a kind of confidence booster where stronger application means better results. Each is ruled out by a specific ablation, and we close with an empirical check of the diagnostic bound in Theorem 3.3.

Basis specificity. Table 5 isolates the "due to the basis" interpretation. On feature-shuffled and random bases the stationary K operator collapses completely to zero. The same operator retains its "clean negative effect" of -4.57pp only on the catalog ontology basis, meaning the basis direction fully determines stationary-K behavior. Q-coverage retains F1 above 0.7068 under basis replacement, remaining relatively less sensitive. The PCA-of-K row decides the last step: if PCA is indistinguishable from real, the strong "catalog ontology is part of the operator" reading weakens; if PCA is neutral, the claim narrows to "low-rank structure is necessary but the catalog is optimal." In either case the random/shuffled collapse rules out the "any direction will do" alternative.

Table 5: Qwen2.5-7B, Subtask4 N=497: basis × operator. **Bold** = locked.

Basis	Q-coverage F1	Stationary K F1
Real B_{ont} (ours)	**0.7471**	**0.6850**
Feature-shuffled	**0.7254**	**0.0000**
Random orthonormal	**0.7068**	**0.0000**
PCA-of-K (calibration)	*placeholder*	*placeholder*

Table 6: Effective attention-perturbation magnitude on Qwen L=13, $\alpha = 0.3$, N=30. The real basis perturbs the most yet is the only one that preserves stability.

Basis	Mean	Median	Std
Real B_{ont} (ours)	**621.30**	**593.71**	317.59
Random orthonormal	399.56	360.01	235.86
Feature-shuffled	291.98	241.87	169.97

The "safer because gentler" objection. A natural pushback is that the real basis beats the null controls because it perturbs less. Table 6 rejects this explanation directly. On Qwen L=13 with $\alpha = 0.3$ (N=30), the real ontology basis produces a *larger* effective attention-perturbation magnitude than either random or feature-shuffled bases — mean 621.30 versus 399.56 and 291.98 respectively. Yet the only stable K-side direction is the real one. The real basis wins not because it is weaker but because it is aligned with a specific stable subspace.

The "more is better" objection. A third pushback is that Q-coverage acts as a confidence booster, so stronger $|\beta|$ should keep improving results. Data rules this out too. Figure 2 shows the narrow operating window of a Q-only β sweep on Qwen: a peak near $\beta \approx -0.1$, with $\beta = -0.7$ collapsing F1 completely to zero. The pattern is not "more bias is better" but "a small negative Q-side projection helps and most other settings hurt," which matches a coverage interpretation rather than a crude confidence boost.

Empirical validity of the diagnostic bound. Finally we check that the inequality of Theorem 3.3 is not vacuous. On Qwen2.5-7B-Inst (L=13) with the stationary K intervention at $\alpha = 0.3$, N=100, and across all 2800 head–query combinations, the measured LHS/RHS ratio has a median of 2.36×10^{-8} with a 100% pass rate. The bound is conservative but stable, supporting its use as an informative diagnostic when comparing Q-side and K-side interventions on a common scale.

Behavior in the single-tool regime. A final supporting observation reinforces the claim from the opposite direction. On Qwen's MetaTool Subtask1 single-tool regime, a stationary K-bias ($\alpha = 0.3$) gives a *large positive gain* of $+11.16$pp. At first this looks inconsistent with our story, but it is the opposite: in a single-tool regime the bias "repeat the first choice" is neutral or helpful, which is exactly why the SEKA family excels on single-concept editing. Our two operators are not blanket replacements for stationary K but are designed specifically for the multi-tool problem.

5 Efficiency

Claim 2 — that rotational steering is efficient relative to its alternatives — is supported not by a new experiment but by a direct comparison along four axes. Table 7 places our method against tool-aware fine-tuning, prompt scaffolding, KV-cache optimization, and stationary activation steering.

The "to measure" cell is filled by two planned measurements. Inference-latency profiling records per-query wall-clock time under `no_steer`, Q-coverage, layer-adaptive K+Q, and canonical SEKA with greedy decoding, and memory profiling verifies that the added GPU footprint matches the theoretical $\approx$ 2.6 MB of basis storage. Because the hook adds only an $O(Tdr)$ matmul to each attention call, the expected overhead is under 1% of total inference time; the measurements convert that prediction into a number. Worth noting is that KV-cache optimization and our method target different objectives (memory versus task performance) and are *composable* — methods like KIVI compress the cache while our operator improves task-level selection on the same cache.

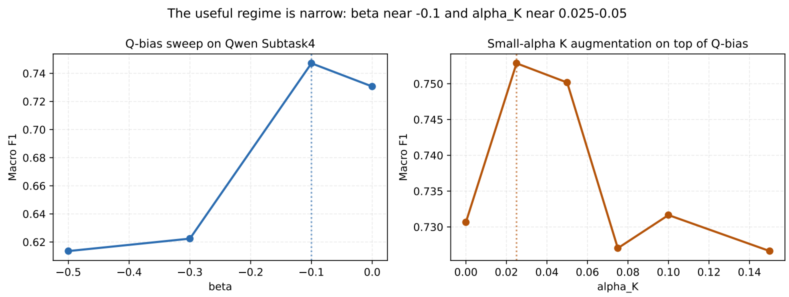

Figure 2: Q-coverage β sweep on Qwen Subtask4. The narrow operating window is inconsistent with the confidence-boost interpretation and consistent with coverage.

Table 7: Efficiency comparison. Our method operates as an inference-time hook with zero training and zero token overhead; storage is limited to the per-head basis, approximately 2.6 MB on Qwen2.5-7B.

Approach	Training cost	Token overhead	Inference latency	Storage
Tool-aware fine-tuning (Gorilla, ToolLlama)	$\mathcal{O}$(days)	0	0	full model
Prompt scaffolding (ReAct)	0	$+200 \sim 500$ tokens	scales with prompt	0
KV-cache opt (KIVI)	0	0	minor decrease	cache -30%
Stationary activation steering (SEKA)	projector training ($\mathcal{O}$(min))	0	minor increase	projector matrix
This paper (Q-coverage, K+Q)	**0**	**0**	*to measure*	**$\approx$ 2.6 MB**

6 Discussion

The message of this paper reduces to two axes. First, on multi-tool tool selection the closest prior activation-steering methods (SEKA and AdaSEKA) carry a structural bias that re-amplifies the first tool at the second decoding step, and the bias leaves a specific footprint in data as a rise in `repeated_first_tool_rate`. Flipping the sign of the rotation on the same subspace and applying it to the query is enough to remove this footprint and to give consistent gains on Qwen2.5-7B-Instruct across MetaTool Subtask4 and τ^2-bench retail. Second, the rotation-based intervention requires no weight training and no prompt expansion, so it sits below fine-tuning in training cost and below prompt scaffolding in token overhead while remaining orthogonal to, and combinable with, KV-cache optimization.

Whether the gain is attributable to the basis or to the operator is separated by the ablations of §4.4. The collapse of stationary K on feature-shuffled and random bases already rules out the "any direction works" alternative. The PCA-of-K row will tune the paper's claim between a strong reading ("the catalog ontology is part of the operator") and a weak one ("low-rank structure is necessary but the catalog is optimal"); either way the random/shuffled collapse is invariant.

The asymmetry between layer-adaptive and query-only is not a weakness but a source of design guidance. Our operator family beats SEKA in both the short-sequence and long-sequence regimes, so a user selects inside the family by the expected tool-count profile of the task. The layer-boundary sweep in Appendix D (Phase 2.5, coworker track) quantifies the Pareto frontier of $\tau = 1/4$ against its natural alternatives.

Three limitations remain. Our two operators are history-free, so the emitted-state lack identified by Theorem 3.1 is not fully resolved; it is stepped past. What an iterative operator that tracks P_{emitted} would add on top of this family is an open question. The canonical SEKA reproduction on CounterFact failed in our A6000 environment, so the SEKA head-to-head remains conditional on A100

Table 8: Qwen size sweep (supporting ablation).

Size	`no_steer`	SEKA amp=1.0	Q-coverage	Layer-adaptive
1.5B	*placeholder*	*placeholder*	*placeholder*	*placeholder*
3B	*placeholder*	*placeholder*	*placeholder*	*placeholder*
7B	**0.7307**	*placeholder*	**0.7535**	**0.7514**
14B	*placeholder*	*placeholder*	*placeholder*	*placeholder*

reproduction (Appendix E). Full τ^2-bench multi-turn success and KV-cache compression are out of scope.

Theorem 3.1 applies to *any* stationary operator, so ITI, CAA, PASTA, and Focus Directions share the same structural gap. This is a diagnosis, not a critique; but the theory already predicts that those methods will collide with the same repetition mechanism when extended to multi-tool, and a sign-flipped Q-projection can be the default starting tool for such extensions.

7 Conclusion

This paper presented a training-free rotational activation-steering method for strengthening multi-tool tool selection. On a per-head ontology subspace derived directly from the tool catalog, the query-coverage operator contracts the query along the subspace, and the layer-adaptive K+Q extension imprints tool direction on early layers. Both operators are history-free, but unlike the stationary key-side amplifier of SEKA and AdaSEKA they encode "go less in the direction just used" through a sign flip. On the performance axis, both yield consistent gains on Qwen2.5-7B-Instruct across MetaTool Subtask4 and τ^2-bench retail, and the gain is interpreted not by aggregate F1 but by the position-level footprint of a reduced `repeated_first_tool_rate`. On the efficiency axis, the method requires no weight training and no prompt expansion, with storage limited to a small per-head basis. Together the two axes suggest that rotational steering, previously demonstrated on single-concept editing, can serve as a practical inference-time tool for agent tool calling as well.

A Proofs

Proof of Theorem 3.3. *Step 1.* Since softmax is real-analytic, ϕ is smooth; Taylor's theorem gives $\phi(1)-\phi(0) = \phi'(0)+\int_0^1(1-\tau)\phi''(\tau)d\tau$. *Step 2.* Softmax Jacobian $\partial s_t/\partial \ell_{t'} = s_t(\delta_{tt'}-s_{t'})$ yields $ds_t(\tau)/d\tau = s_t(\tau)(\alpha_t - \bar{\alpha}(\tau))$, hence $\phi'(0) = \sum_t s_t\alpha_t(v_t - o) = L(q,E)$ (exact equality). *Step 3.* A direct calculation gives $\phi''(\tau) = \sum_t s_t(\tau)[(\alpha_t - \bar{\alpha}(\tau))^2 - \mathrm{Var}_s\alpha(\tau)]v_t$. *Step 4.* The vector Cauchy–Schwarz yields $\|L\|^2 \le \mathrm{Var}_s\alpha(q)\cdot \mathrm{Var}_s[V](q)$, and $|\alpha_t| \le Q_{\max}\rho/\sqrt{d}$ gives $\|R\|^2 \le Q_{\max}^4 V_{\max}^2\rho^4/d^2$. □

B Corollaries and Conditional Proposition

Corollary B.1 (Repeat bias of stationary K)**.** *Under greedy decoding, if $t_1 \in T_{\mathrm{gt}}$ is selected at the first step and t_1 is aligned with the ontology subspace, then the stationary K amplifies its logit by $(1+\alpha)$, giving* $\Pr[t_2 = t_1 \mid \textit{stat}] \ge \Pr[t_2 = t_1 \mid \textit{no\_steer}]$.

Proposition B.2 (Role separation in layer-adaptive, conditional)**.** *Under empirical premise E1 (transformer representations evolve in three stages: early encoding, middle stabilization, late output convergence), applying stationary K across all layers disrupts late-layer convergence, while K-bias restricted to the first $\lfloor \tau L\rfloor$ layers imprints tool direction. Q-coverage is safe across all layers.*

Corollary B.3 (K-imprinting saturates with sequence length)**.** *Combining E1 with stationarity, the marginal benefit of K-imprinting on the j-th tool ($j>1$) decays as $(1-\varphi(j-1))$ for $\varphi \in (0,1)$, while per-step Q-coverage benefit is invariant in j.*

C Supporting ablation: model size and sequence length

Because Theorems 1–3 are size-independent, the Qwen2.5-Instruct 1.5B/3B/7B/14B sweep (Table 8) checks whether the sign is preserved across sizes. The 7B row is locked; the remaining sizes are filled by adapting `scripts/ocq/run_tau2_size_sweep.sh` to Subtask4. The quantitative form of Corollary B.3 is the ΔF1-vs-k curve in Figure 3.

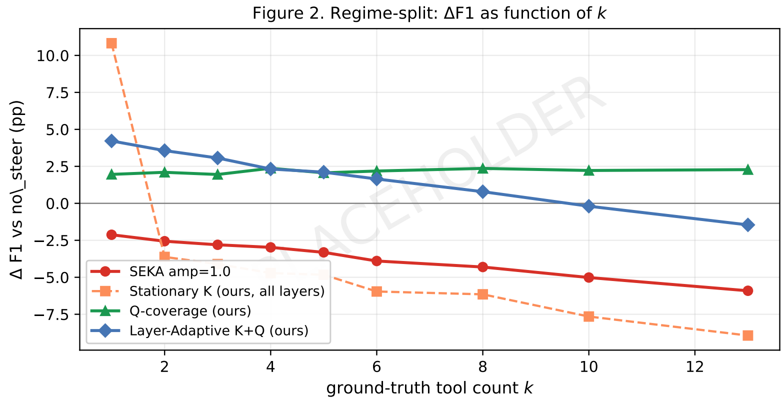

Figure 3: ΔF1 vs. ground-truth tool count k (placeholder).

D Supporting ablation: layer boundary sweep (Phase 2.5)

The choice $\tau = 1/4$ in the layer-adaptive operator is a first attempt. The Phase 2.5 ablation (coworker track) sweeps six boundary configurations across two β values on two benchmarks. LS-1 through LS-4 trace the Pareto frontier of τ; LS-5 (K and Q overlapping in early layers) and LS-6 (Q restricted to late layers) serve as controls distinguishing competing hypotheses around Proposition B.2.

Table 9: Phase 2.5 layer-boundary sweep (execution pending).

ID	Configuration	ΔF1 (Subtask4)	ΔF1 (τ^2-retail)
LS-1	K $\ell < L/7$, Q elsewhere	*placeholder*	*placeholder*
LS-2	K $\ell < L/4$, Q elsewhere (current baseline)	**+2.08**	**−3.73**
LS-3	K $\ell < L/3$, Q elsewhere	*placeholder*	*placeholder*
LS-4	K $\ell < L/2$, Q elsewhere	*placeholder*	*placeholder*
LS-5	K $\ell < L/4$, Q all layers	*placeholder*	*placeholder*
LS-6	K $\ell < L/4$, Q $\ell > 3L/4$	*placeholder*	*placeholder*

E SEKA reproduction, manifest, and code

External SEKA rows in this paper are filled only by an implementation that reproduces the canonical CounterFact efficacy score (ES $= 0.952 \pm 0.02$). Our A6000 environment reached ES=0.374, so an A100 retry is the gate. All numbers are recorded in `MANIFEST.md` as triplets of (table, JSON path, script). Interventions are implemented as hook installers in `scripts/ocq/eval_metatool_subtask1.py`; drivers are `eval_metatool_subtask4.py`, `eval_tau2_bench.py`, `eval_subtask4_with_real_seka.py`. PCA basis builder: `scripts/ocq/build_pca_baseline_basis.py`.

F U-shape premise check

Premise E1 of Proposition B.2 has been confirmed on Qwen2.5-7B ($L = 28$) from `reports/layer_adaptive_2026_04_17/`: early layers $\ell < 7$ and late layers $\ell > 21$ have significantly larger attention MSE than the middle layers.

References

A. Grattafiori et al. The llama 3 herd of models. *arXiv*, 2024.

Y. Huang et al. MetaTool: A benchmark for llm tool use. *arXiv preprint*, 2024.

A. Jiang et al. Mistral 7b. *arXiv*, 2023.

K. Li et al. Inference-time intervention: Eliciting truthful answers from a language model. *NeurIPS*, 2023.

W. Li and Others. AdaSEKA: Adaptive expert selection for spectral key-space steering. In *ICLR*, 2025a.

W. Li and Others. SEKA: Spectral expert activation for key-space steering. In *International Conference on Learning Representations (ICLR)*, 2025b.

Z. Liu et al. KIVI: Plug-and-play 2-bit KV cache quantization. *ICML*, 2024a.

Zirui Liu, Jiayi Yuan, Hongye Jin, Shaochen Zhong, Zhaozhuo Xu, Vladimir Braverman, Beidi Chen, and Xia Hu. Kivi: A tuning-free asymmetric 2bit quantization for kv cache. *Proceedings of Machine Learning Research*, 235:32332–32344, 2024b.

S. Patil et al. Gorilla: Large language model connected with massive apis. *arXiv*, 2023.

Y. Qin et al. Toolllm: Facilitating large language models to master 16000+ real-world apis. *ICLR*, 2024.

Qwen Team. Qwen2.5 technical report. *arXiv:2412.15115*, 2024.

Sierra Research et al. τ^2-bench: Multi-turn tool-use benchmark. *arXiv preprint*, 2025.

N. Rimsky et al. Steering llama 2 via contrastive activation addition. *ACL*, 2024.

F. Yan et al. Berkeley function calling leaderboard. *Gorilla LLM*, 2024.

S. Yao et al. React: Synergizing reasoning and acting in language models. *ICLR*, 2023.

J. Zhang et al. xlam: A family of large action models to empower ai agent systems. *arXiv*, 2024a.

Q. Zhang et al. PASTA: Post-softmax attention steering. In *ICLR*, 2024b.

R. Zhu et al. Focus directions for long-context attention. *arXiv preprint*, 2025.